UNIT 6 - COMPLETE NOTES

Algebraic Manipulation

Highest common factor and least common multiple of algebraic expressions, basic operations on algebraic fractions, and the square root of an algebraic expression.

What this unit covers

- **6.1** Highest Common Factor (HCF) - by factorisation and by division
- **6.1** Least Common Multiple (LCM) - by factorisation and by formula
- **6.1** Relationship between HCF and LCM, and its applications
- **6.2** Basic operations on algebraic fractions - add, subtract, multiply, divide
- **6.3** Square root of algebraic expressions - by factorisation and by division

How to use these notes

Read one section at a time and work through every worked example with pen and paper before looking at the solution. Yellow boxes hold the rules most often tested in board papers. The last two pages give solved examples, board-style multiple-choice questions and short questions for self-testing.

Exam focus: Board papers regularly ask for (1) the HCF of two or three expressions by factorisation, (2) the HCF by the division method, (3) the LCM using the formula LCM = product / HCF, (4) finding a missing polynomial from a given HCF and LCM, (5) simplifying an algebraic fraction, and (6) the square root of an expression by division.

Based on *A Textbook of Mathematics for Grade IX*, Khyber Pakhtunkhwa Textbook Board (KPTB), Peshawar - Unit 6, starting at book page 149. Notes prepared by Shah Fahad.

1. Highest Common Factor (HCF)

Definition - HCF: The Highest Common Factor of two or more polynomials is the polynomial of **highest degree** that divides each of them exactly. For monomials, also take the HCF of the coefficients and the **lowest** power of each common variable.

HCF by factorisation - the method

1. Factorise every given expression completely.
2. Pick out the factors that appear in **every** expression.
3. Where a factor repeats, take it with its **lowest** power.
4. Multiply these common factors together - that product is the HCF.

Example 1 (book). Find the HCF of $x^2 - y^2$ and $x^2 - xy$.
Solution: $x^2 - y^2 = (x + y)(x - y)$ and $x^2 - xy = x(x - y)$. The only common factor is $(x - y)$, so **HCF = x - y**.

Example 2. Find the HCF of $12x^2y$ and $18xy^2$.
Solution: HCF of the coefficients 12 and 18 is 6. Lowest powers of the common variables: x^1 and y^1. So **HCF = 6xy**.

Example 3 (book). Find the HCF of $ax^2 + 5ax + 6a$, $ax^3 + 9ax^2 + 14ax$ and $15a(x^2 - 4)$.
Solution: Factorising each one:
$ax^2 + 5ax + 6a = a(x + 3)(x + 2)$
$ax^3 + 9ax^2 + 14ax = ax(x + 7)(x + 2)$
$15a(x^2 - 4) = 15a(x - 2)(x + 2)$
The factors common to all three are a and $(x + 2)$, so **HCF = a(x + 2)**.

Remember: For the HCF take common factors only, with the **lowest** powers. If no factor (other than 1) is common to all the expressions, the HCF is 1.

2. HCF by the division method

When expressions are hard to factorise, use repeated division. This is the same idea as the division method for numbers.

Steps

1. Write each polynomial in **descending order** of the variable.
2. Divide the higher-degree polynomial by the lower-degree one. If the degrees are equal, divide either by the other.
3. Now divide the **divisor** by the **remainder**.
4. Repeat until the remainder is **zero**.
5. The **last divisor** is the required HCF. A constant factor in a remainder may be dropped - it does not change the HCF.

> **Three polynomials:** First find the HCF of any two of them, then find the HCF of that result and the third polynomial.

Example 4 (book). Find the HCF of $2x^3 + 7x^2 + 4x - 4$ and $2x^3 + 9x^2 + 11x + 2$ by division.
Solution: Call them P and Q. Both have degree 3, so divide Q by P:

Step 1: Q divided by P gives quotient 1 and remainder $\mathbf{2x^2 + 7x + 6}$.

Step 2: P divided by $(2x^2 + 7x + 6)$ gives quotient x and remainder $-2x - 4 = -2(x + 2)$. Dropping the constant -2, the new divisor is $\mathbf{x + 2}$.

Step 3: $(2x^2 + 7x + 6)$ divided by $(x + 2)$ gives quotient $2x + 3$ and remainder **0**.

The last divisor is the HCF, so **HCF = x + 2**. Check: $P = (x + 2)(2x^2 + 3x - 2)$ and $Q = (x + 2)(2x^2 + 5x + 1)$.

Example 5 (book). Find the HCF of $x^3 - 6x^2 + 11x - 6$, $3x^3 - 5x^2 + 6x - 4$ and $3x^3 + 5x^2 - 6x - 2$.
Solution: Applying the division method to the last two polynomials gives HCF = x - 1. Dividing the first polynomial by (x - 1) leaves remainder 0, so x - 1 also divides it exactly. Hence the HCF of all three is **x - 1**.

> **Watch out:** Arrange every polynomial in descending order before you divide, and fill any missing power with a 0 term in your working. Most lost marks in this topic come from sign errors during subtraction.

3. Least Common Multiple (LCM)

Definition - LCM: The Least Common Multiple of two or more polynomials is the polynomial of **lowest degree** that is divisible by each of the given polynomials.

LCM by factorisation - the method

$$\text{LCM} = \text{Common factors} \times \text{Non-common factors}$$

Factorise every expression. Take each common factor **once**, with its **highest** power, and multiply by all the remaining (non-common) factors. For monomials, also take the LCM of the coefficients.

Example 6. Find the LCM of $6x^2$ and $9xy$.
Solution: LCM of the coefficients 6 and 9 is 18. Highest powers: x^2 and y^1. So **LCM = $18x^2y$**.

Example 7 (book). Find the LCM of $x^2 + 4x + 4$ and $x^2 + 5x + 6$.
Solution: $x^2 + 4x + 4 = (x + 2)^2$ and $x^2 + 5x + 6 = (x + 3)(x + 2)$. The common factor $(x + 2)$ is taken with its highest power, $(x + 2)^2$, and the non-common factor is $(x + 3)$. So **LCM = $(x + 2)^2(x + 3)$**.

Example 8 (book). Find the LCM of $x^2 - 4x + 3$, $x^2 - 3x + 2$ and $x^2 - 5x + 6$.
Solution: Factorising: $(x - 3)(x - 1)$, $(x - 2)(x - 1)$ and $(x - 3)(x - 2)$. Each of $(x - 1)$, $(x - 2)$, $(x - 3)$ is common to one pair, and there are no other factors. So **LCM = $(x - 1)(x - 2)(x - 3)$**.

HCF or LCM? HCF: common factors, **lowest** powers - the answer divides every expression. LCM: common factors once with **highest** powers, times all the rest - every expression divides the answer. Mixing these two up is the most common error in this unit.

4. The HCF-LCM relationship

For **two** polynomials A and B, with HCF = H and LCM = L, the book proves the following relationship (book section 6.1.3). Write A = H x a and B = H x b, where a and b have no common factor. Then L = H x a x b, so H x L = (H x a) x (H x b) = A x B.

$$A \times B = H \times L \quad \text{i.e.} \quad \text{Product of two polynomials} = \text{HCF} \times \text{LCM}$$

$$\text{LCM} = (\text{Product of two polynomials}) / \text{HCF}$$

Example 9. Verify the relationship for $A = x^2 - 1$ and $B = x^2 - 3x + 2$.
Solution: $A = (x - 1)(x + 1)$ and $B = (x - 1)(x - 2)$, so HCF = $x - 1$ and LCM = $(x - 1)(x + 1)(x - 2)$. Then HCF x LCM = $(x - 1)^2(x + 1)(x - 2)$, which is exactly A x B. The relationship holds.

Example 10 (book). The HCF and LCM of two polynomials are $x - 3$ and $x^3 - 9x^2 + 26x - 24$. Find the second polynomial when one polynomial is $x^2 - 5x + 6$.
Solution: Second polynomial B = (H x L) / A = $(x - 3)(x^3 - 9x^2 + 26x - 24) / (x^2 - 5x + 6)$. Dividing the LCM by A: $(x^3 - 9x^2 + 26x - 24) / (x^2 - 5x + 6) = x - 4$. So B = $(x - 3)(x - 4) = \mathbf{x^2 - 7x + 12}$.

Example 11 (book). The HCF of two polynomials of degree 2 is $x - 1$ and their LCM is $x^3 + 4x^2 + x - 6$. Find the polynomials.
Solution: Factorise the LCM: $x^3 + 4x^2 + x - 6 = (x - 1)(x + 2)(x + 3)$. The common factor is $(x - 1)$; the non-common factors $(x + 2)$ and $(x + 3)$ go one to each polynomial. So the polynomials are $(x - 1)(x + 2) = \mathbf{x^2 + x - 2}$ and $(x - 1)(x + 3) = \mathbf{x^2 + 2x - 3}$.

Real-life use (book): The sum of two numbers is 120 and their HCF is 12. Writing the numbers as 12x and 12y gives x + y = 10 with x, y having no common factor. The pairs are (1, 9) and (3, 7), so the numbers are **12 and 108**, or **36 and 84**. Note: the product formula A x B = H x L is for **two** polynomials only - it does not hold for three.

5. Algebraic fractions - addition and subtraction

> **Definition - Algebraic fraction:** An algebraic fraction is the quotient of two algebraic expressions, such as $(3x^2 + 4a) / (x + 1)$. Throughout this section, all fractions are assumed to be defined - no denominator is zero.

The method

1. **Factorise** every denominator first.
2. Take the **LCM of the denominators** as the common denominator.
3. Multiply each numerator by whatever its denominator was multiplied by.
4. Add or subtract the numerators, then **simplify and cancel** any common factor.

Example 12 (book). Simplify $(x + y) / (3x + 2y) + (x - y) / (3x + 2y)$.
Solution: The denominators are already the same, so add the numerators: $[(x + y) + (x - y)] / (3x + 2y) =$ **$2x / (3x + 2y)$**.

Example 13. Simplify $1 / (x - 2) + 1 / (x + 2)$.
Solution: The LCM of the denominators is $(x - 2)(x + 2) = x^2 - 4$. So the sum $= [(x + 2) + (x - 2)] / (x^2 - 4) =$ **$2x / (x^2 - 4)$**.

Example 14 (book). Simplify $(x - y) / (x + y) - (x^2 - 2y^2) / (x^2 - y^2)$.
Solution: Since $x^2 - y^2 = (x + y)(x - y)$, the common denominator is $x^2 - y^2$. The numerator becomes $(x - y)^2 - (x^2 - 2y^2) = x^2 - 2xy + y^2 - x^2 + 2y^2 = 3y^2 - 2xy$. Answer: **$(3y^2 - 2xy) / (x^2 - y^2)$**.

> **Watch out:** In subtraction, the minus sign applies to the **whole** second numerator - put it in brackets first, as in Example 14, where $-(x^2 - 2y^2)$ became $-x^2 + 2y^2$. Never cancel a term that is only part of a sum: factors cancel, terms do not.

6. Algebraic fractions - multiplication and division

The method

- **Multiply:** factorise all numerators and denominators, cancel common factors, then multiply what is left - numerators together, denominators together.
- **Divide:** turn the second fraction upside down (take its reciprocal) and multiply. Division questions are multiplication questions in disguise.
- **Reduce first:** a single fraction is simplified by cancelling a factor common to its numerator and denominator, using the HCF.

Example 15 (book). Simplify $[(x + 4) / (x - 3)] \times [(x^2 - 9) / (x^2 - x - 2)]$.
Solution: Factorise: $x^2 - 9 = (x + 3)(x - 3)$ and $x^2 - x - 2 = (x - 2)(x + 1)$. Cancel the common factor $(x - 3)$. Answer: **$(x + 4)(x + 3) / [(x - 2)(x + 1)]$**.

Example 16 (book). Simplify $[3 / (x - 2) - 1 / (x + 1)]$ divided by $(x + 4) / (x - 2)$.
Solution: Inside the bracket, the common denominator is $(x - 2)(x + 1)$: $[3(x + 1) - (x - 2)] / [(x - 2)(x + 1)] = (2x + 5) / [(x - 2)(x + 1)]$. Now divide by multiplying by the reciprocal $(x - 2) / (x + 4)$ and cancel $(x - 2)$. Answer: **$(2x + 5) / [(x + 1)(x + 4)]$**.

Example 17 (book). Simplify $y / (y^2 - y - 2) - 1 / (y^2 + 5y - 14) - 2 / (y^2 + 8y + 7)$.
Solution: Factorise the denominators: $(y + 1)(y - 2)$, $(y - 2)(y + 7)$ and $(y + 1)(y + 7)$. Their LCM is $(y + 1)(y - 2)(y + 7)$. The numerator becomes $y(y + 7) - (y + 1) - 2(y - 2) = y^2 + 4y + 3 = (y + 1)(y + 3)$. Cancelling $(y + 1)$ gives **$(y + 3) / [(y - 2)(y + 7)]$**.

> **Order of work:** Factorise first, cancel second, multiply last. Students who multiply everything out first create degree-4 expressions that are slow and error-prone to simplify.

7. Square root by factorisation

The square root of an algebraic expression is found in two ways: **(a) by factorisation** and **(b) by division**. In the factorisation method, first rewrite the expression as a **perfect square**, usually with one of the identities $(a + b)^2 = a^2 + 2ab + b^2$ or $(a - b)^2 = a^2 - 2ab + b^2$, then take the square root. Remember that a square root has two values, written with the **plus-minus** sign.

Example 18 (book). Find the square root of $x^2 + ax + (1/4)a^2$ by factorisation.
Solution: $x^2 + ax + (1/4)a^2 = x^2 + 2(x)(a/2) + (a/2)^2 = (x + a/2)^2$. So the square root is **+/-(x + a/2)**.

Example 19. Find the square root of $4a^2 - 12ab + 9b^2$.
Solution: $4a^2 - 12ab + 9b^2 = (2a)^2 - 2(2a)(3b) + (3b)^2 = (2a - 3b)^2$. So the square root is **+/- (2a - 3b)**.

Example 20. Find the square root of $x^4 + 2x^3 + 3x^2 + 2x + 1$.
Solution: Group the terms as a square of a trinomial: $(x^2 + x + 1)^2 = x^4 + 2x^3 + (1 + 2)x^2 + 2x + 1 = x^4 + 2x^3 + 3x^2 + 2x + 1$. So the square root is **+/- (x^2 + x + 1)**.

Example 21 (book). Find the square root of $x^2 + 1/x^2 - 10(x + 1/x) + 27$.
Solution: Write 27 as 25 + 2. Since $x^2 + 1/x^2 + 2 = (x + 1/x)^2$, the expression becomes $(x + 1/x)^2 - 10(x + 1/x) + 25 = [(x + 1/x) - 5]^2$. So the square root is **+/- (x + 1/x - 5)**.

How to spot the square: Check the first and last terms - both should be perfect squares. Then check that the middle term equals twice the product of their square roots (with the sign of the middle term). If it does, the expression is (root of first +/- root of last)^2.

8. Square root by the division method

Use division when the expression does not obviously form a perfect square pattern. The rules below are the book's rules, shown on its own example.

Rules

1. Write the expression in **descending order** of the variable.
2. Take the square root of the **first term** - this is the first term of the answer. Subtract its square from the expression.
3. Find the next term of the root: divide the first term of the remainder by **twice the root found so far**.
4. Multiply that new term by (twice the root found so far **plus** the new term) and subtract from the remainder.
5. Repeat until the remainder is **zero**.

Example 22 (book). Find the square root of $16x^4 - 24x^3 + 25x^2 - 12x + 4$ by division.
Solution:
Term 1: sqrt($16x^4$) = $4x^2$. Subtract $(4x^2)^2 = 16x^4$. Remainder: $-24x^3 + 25x^2 - 12x + 4$.
Term 2: $-24x^3$ / (2 x $4x^2$) = $-3x$. Multiply: $-3x(8x^2 - 3x) = -24x^3 + 9x^2$. Subtract. Remainder: $16x^2 - 12x + 4$.
Term 3: $16x^2$ / (2 x $4x^2$) = 2. Multiply: $2(8x^2 - 6x + 2) = 16x^2 - 12x + 4$. Subtract. Remainder: **0**.
So the square root is **+/- ($4x^2 - 3x + 2$)**. Check by squaring: $(4x^2 - 3x + 2)^2$ gives back the original expression exactly.

Example 23. Find the square root of $x^4 + 6x^3 + 13x^2 + 12x + 4$ by division.
Solution: Term 1: sqrt(x^4) = x^2; remainder $6x^3 + 13x^2 + 12x + 4$. Term 2: $6x^3$ / $2x^2$ = $3x$; multiply $3x(2x^2 + 3x) = 6x^3 + 9x^2$; remainder $4x^2 + 12x + 4$. Term 3: $4x^2$ / $2x^2$ = 2; multiply $2(2x^2 + 6x + 2) = 4x^2 + 12x + 4$; remainder 0. Square root = **+/- ($x^2 + 3x + 2$)**.

Always check: Square your final answer. If squaring does not return the original expression, a sign or coefficient is wrong. This one check catches almost every division-method error.

9. Solved examples - mixed practice

Example 24. Find the HCF of $(x + 6)^2$ and $x^2 - 36$. (Book Exercise 6.1)
Solution: $(x + 6)^2 = (x + 6)(x + 6)$ and $x^2 - 36 = (x + 6)(x - 6)$. The common factor is $(x + 6)$, so **HCF = x + 6**.

Example 25. Find the LCM of $x - 3$, $x^2 - 9$ and $(x - 3)^2$.
Solution: $x^2 - 9 = (x - 3)(x + 3)$. Take the common factor $(x - 3)$ with its highest power, $(x - 3)^2$, times the non-common factor $(x + 3)$. **LCM = $(x - 3)^2(x + 3)$**.

Example 26. Simplify $(x^2 - 25) / (5 - x)$.
Solution: $(x^2 - 25) / (5 - x) = (x - 5)(x + 5) / (5 - x)$. Since $(x - 5) = -(5 - x)$, cancelling gives **$-(x + 5)$**. Watch the sign - this is a favourite board trap.

Example 27. Two polynomials have HCF = $x - 1$ and LCM = $(x - 1)(x + 2)(x + 3)$. One polynomial is $x^2 + x - 2$. Find the other.
Solution: Other = (HCF x LCM) / given polynomial = $(x - 1)^2(x + 2)(x + 3) / [(x - 1)(x + 2)] = (x - 1)(x + 3)$ = **$x^2 + 2x - 3$**.

Example 28. Is $(3x^2 - x + 4)$ the square root of $9x^4 - 6x^3 + 25x^2 - 8x + 16$? Check by squaring.
Solution: Square it: $(3x^2 - x + 4)^2 = 9x^4 + x^2 + 16 - 6x^3 + 24x^2 - 8x = 9x^4 - 6x^3 + 25x^2 - 8x + 16$. Yes - it matches exactly, so the square root is **+/- $(3x^2 - x + 4)$**.

Common mistakes to avoid

- Taking the **highest** powers for the HCF or the **lowest** powers for the LCM - it is the other way round.
- Using A x B = HCF x LCM for **three** polynomials. The formula is for two only.
- Cancelling terms instead of factors in a fraction, for example cancelling the x in $(x + 3) / x$. Only a factor of the **whole** numerator and denominator can cancel.
- Forgetting the **+/-** sign on a square root, and forgetting the sign change in $(x - 5) / (5 - x) = -1$.
- In the division method for HCF, stopping at the first non-zero remainder. Continue until the remainder is exactly zero - the **last divisor** is the answer.

10. Test yourself - MCQs and short questions

Multiple-choice questions

1. The HCF of $x^2 - y^2$ and $x^2 - xy$ is: (a) $x + y$ (b) $x - y$ (c) x (d) $x(x - y)$
   Answer: (b)
2. The HCF of $12x^2y$ and $18xy^2$ is: (a) $36x^2y^2$ (b) $6xy$ (c) $6x^2y^2$ (d) $3xy$
   Answer: (b)
3. The LCM of $x^2 + 4x + 4$ and $x^2 + 5x + 6$ is: (a) $(x + 2)(x + 3)$ (b) $(x + 2)^2(x + 3)$ (c) $(x + 2)^2$ (d) $(x + 3)^2(x + 2)$
   Answer: (b)
4. HCF = $x - 3$, LCM = $x^3 - 9x^2 + 26x - 24$, one polynomial = $x^2 - 5x + 6$. The other is: (a) $x^2 - 7x + 12$ (b) $x^2 + 7x + 12$ (c) $x - 4$ (d) $x^2 - 5x + 6$
   Answer: (a)
5. $(x + y) / (3x + 2y) + (x - y) / (3x + 2y)$ equals: (a) $2y / (3x + 2y)$ (b) $2x / (3x + 2y)$ (c) 1 (d) $(x + y) / (3x + 2y)$
   Answer: (b)
6. $(x^2 - 25) / (5 - x)$ simplifies to: (a) $x + 5$ (b) $x - 5$ (c) $-(x + 5)$ (d) $5 - x$
   Answer: (c)
7. The square root of $16x^4 - 24x^3 + 25x^2 - 12x + 4$ is: (a) $4x^2 - 3x + 2$ (b) $4x^2 + 3x + 2$ (c) $16x^2 - 3x + 2$ (d) $4x^2 - 3x - 2$
   Answer: (a)
8. If HCF = H and LCM = L for two polynomials A and B, then: (a) A + B = H + L (b) A x B = H x L (c) A / B = H / L (d) A x B = H / L
   Answer: (b)

Short questions for written practice

1. Find the LCM of $x - 3$, $x^2 - 9$ and $(x - 3)^2$. (Answer: $(x - 3)^2(x + 3)$)
2. Simplify $3 / (y - 2) - 2 / (y + 2) - y / (y^2 - 4)$. (Answer: $10 / (y^2 - 4)$)
3. Simplify $[(x^2 + 5x + 4) / 4y^3] \times [2y^2 / (x^2 + 3x + 2)]$. (Answer: $(x + 4) / [2y(x + 2)]$)
4. Find the square root of $x^4 + 2x^3 + 3x^2 + 2x + 1$ by factorisation. (Answer: $+/- (x^2 + x + 1)$)
5. HCF = $x - 1$, LCM = $x^3 + 4x^2 + x - 6$, one polynomial = $x^2 + x - 2$. Find the other. (Answer: $x^2 + 2x - 3$)

> **One-page revision:** HCF - common factors, lowest powers; by division, the last divisor wins. LCM - common factors once, highest powers, times the rest. A x B = HCF x LCM (two polynomials). Fractions - LCM of denominators, cancel factors only. Square root - perfect square or division; check by squaring.